

Mathematics for Computing

Topic: Sets

Pavithra C J

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DCS MTECH CUSAT

Sets: Definition and Representation

Definition: A set is a well-defined collection of distinct objects.

Roster Form: List all elements

Ex: $A = \{2, 4, 6, 8\}$

Set Builder Form: Describe property

Ex: $A = \{x | x \text{ is an even number and } x \leq 8\}$

AI/ML Example:

Roster: $F = \{\text{age, income, gender}\}$

Set Builder: $F = \{x | x \text{ is a feature used in training}\}$

10 Examples: Roster + Set Builder Form

1. $A = \{1, 2, 3, 4, 5\} = \{x | x \in \mathbb{N}, 1 \leq x \leq 5\}$
2. $B = \{a, e, i, o, u\} = \{x | x \text{ is a vowel}\}$
3. $C = \{2, 4, 6, 8, 10\} = \{x | x \text{ is even}, x \leq 10\}$
4. $D = \{3, 5, 7, 11, 13\} = \{x | x \text{ is prime}, x < 15\}$
5. $E = \{\text{Mon, Tue, ..., Sun}\} = \{x | x \text{ is a day of week}\}$
6. $F = \{0, 1\} = \{x | x \text{ is a binary digit}\}$ - Used in ML
7. $G = \{\text{Python, R, Julia}\} = \{x | x \text{ is a language for AI}\}$
8. $H = \{\text{Accuracy, Precision, Recall}\} =$
 $\{x | x \text{ is a classification metric}\}$
9. $I = \{\text{CNN, RNN, Transformer}\} = \{x | x \text{ is a DL architecture}\}$
10. $J = \{100, 200, 300\} = \{x | x \text{ is a batch size used in training}\}$

Types of Sets

1. Finite Set

Def: Set with countable number of elements. Cardinality $n(A)$ is finite.

Ex: $A = \{1, 2, 3\}$, $n(A) = 3$

Set Builder: $\{x | x \in \mathbb{N}, x \leq 3\}$

2. Infinite Set

Def: Set with uncountable elements. Cardinality is not finite.

Ex: $\mathbb{N} = \{1, 2, 3, \dots\}$

Set Builder: $\{x | x \in \mathbb{N}\}$

3. Null Set / Empty Set ϕ

Def: Set with no elements. $n(\phi) = 0$

Ex: $\{\}$

Set Builder: $\{x | x^2 = -1, x \in \mathbb{R}\}$

Types of Sets - Continued

4. Singleton Set

Def: Set with exactly 1 element. $n(A) = 1$

Ex: $\{0\}$ - bias in ML

Set Builder: $\{x|x = 0\}$

5. Equal Sets

Def: $A = B$ if they have exactly same elements. Cardinality must be equal.

Ex: $A = \{1, 2\}$, $B = \{2, 1\}$

6. Equivalent Sets

Def: Sets with same cardinality $n(A) = n(B)$ but not necessarily same elements.

Ex: $A = \{a, b\}$, $B = \{1, 2\}$. $n(A) = n(B) = 2$

7. Countable Set

Def: Set whose elements can be put in 1-1 correspondence with $\mathbb{N}$.

Types of Sets - Continued

8. Subset $A \subseteq B$

Def: Every element of A is in B.

Ex: $A = \{1, 2\}$, $B = \{1, 2, 3\}$

Set Builder: $A = \{x | x \in B, x \leq 2\}$

9. Super Set $A \supseteq B$

Def: B is a subset of A.

10. Universal Set U

Def: Set of all elements under discussion.

AI/ML Ex: $U = \{x | x \text{ is all data points in dataset}\}$

11. Power Set $P(A)$

Def: Set of all subsets of A. $n(P(A)) = 2^{n(A)}$

Ex: If $A = \{a, b\}$, $P(A) = \{\phi, \{a\}, \{b\}, \{a, b\}\}$

Set Operations

1. Union $A \cup B = \{x | x \in A \text{ or } x \in B\}$

Ex: $\{1, 2\} \cup \{2, 3\} = \{1, 2, 3\}$

2. Intersection $A \cap B = \{x | x \in A \text{ and } x \in B\}$

3. Difference $A - B = \{x | x \in A \text{ and } x \notin B\}$

4. Symmetric Difference

$A \Delta B = \{x | x \in A \text{ or } x \in B, \text{ but not both}\}$

$= (A - B) \cup (B - A)$

5. Complement $A' = \{x | x \in U \text{ and } x \notin A\}$

6. Cartesian Product $A \times B = \{(a, b) | a \in A, b \in B\}$

AI/ML Ex: Hyperparameter grid search

General Inclusion-Exclusion Principle

Used to find cardinality of union.

General Form:

$$\begin{aligned} n\left(\bigcup_{i=1}^n A_i\right) &= \sum_{i=1}^n n(A_i) - \sum_{i < j} n(A_i \cap A_j) + \sum_{i < j < k} n(A_i \cap A_j \cap A_k) - \dots \\ &\quad + (-1)^{n-1} n(A_1 \cap \dots \cap A_n) \end{aligned}$$

Example for 3 sets:

$$\begin{aligned} n(A \cup B \cup C) &= \\ n(A) + n(B) + n(C) &- n(A \cap B) - n(B \cap C) - n(C \cap A) + n(A \cap B \cap C) \end{aligned}$$

Hasse Diagram: Definition

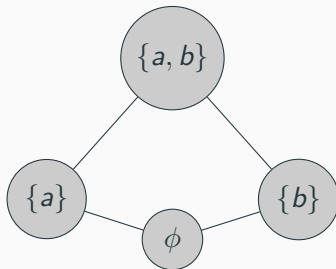
Definition: A Hasse Diagram is a graphical representation of a partially ordered set. It shows the relation using lines and avoids transitive edges.

Rules:

1. Draw smaller elements below larger elements
2. Connect two elements with a line if one covers the other

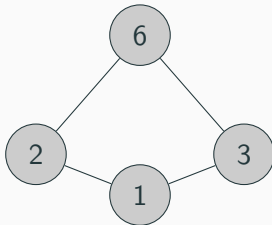
Hasse Diagram: Example 1

Power Set of $\{a, b\}$ ordered by $\subseteq$



Hasse Diagram: Example 2

Set $\{1, 2, 3, 6\}$ ordered by divisibility



Thank You